Sanad Sawalced

COURSE: **CHE 312: Transport Phenomena II**

Spring Semester 2020

Department of Chemical Engineering

University of Illinois at Chicago

Chicago, Illinois 60607-7052

8.0/10

Quiz #3

March 11, 2020

Closed Notes and Book

Duration: 25 minutes

A fluid at 373 K is to be cooled by a different stream of the same fluid at 293 K. A heat exchanger is available for parallel or counter flow. When the heat exchanger is used with parallel flow, the outlet temperature of the hot stream is 338 K. If the mass flow rates in the hot and cold streams are equal to each other and the heat capacity of the fluid is temperature independent, determine:

a) The outlet temperature of the cold stream with parallel flow in the heat exchanger, and

~~b) The outlet temperatures of the cold and hot streams when a counter flow set up is used.~~

a-) $q_1 = \dot{m} C_p \Delta T$ $\quad q_2 = \dot{m} C_p \Delta T$

$\dot{q}_1 = \dot{q}_2$

$\cancel{\dot{m} c_p} \Delta T = \cancel{\dot{m} c_p} \Delta T$

$(373 - 338)k = (T_{out} - 293k)$

$T_{out} = (373 - 338) + (293k)$

$T_{out} = 328 k \longrightarrow T_{out}$ cold.

$$\epsilon = \frac{q_{actual}}{q_{max}} = \frac{373-338}{373-293} = 0.4375$$

parallel flow

$$\epsilon = \frac{1-\exp(-NTU(1+Cr))}{1+Cr} \qquad Cr = 1$$

$$0.4375 = \frac{1-e^{-2NTU}}{2}$$

$$0.875 = 1-e^{-2NTU}$$

$$Ln(0.875) = ln(1) - 2NTU$$

$$-0.13353 = 0 - 2NTU$$

$$NTU = 0.06677$$

for counter flow

$$\epsilon = \frac{NTU}{1+NTU} = \frac{0.06677}{1.06677} = 0.06259$$

$$0.06259 = \frac{338-T_{h,o}}{373-293} \Rightarrow T \Rightarrow (80)(0.06259)+338$$

$$T_{h,o} = 343.01 \text{ K}$$

$$0.06259 = \frac{T_{co}-338}{373-293} \Rightarrow T = 338-(80)(0.06259)$$

$$T_{c,o} = 332.993$$